

FARIZ ZEYNALOV

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EXPERIENCE

Regional NBCU

Credit Risk ML Engineer

Remote (Baku, Azerbaijan)

July 2025 – Present

- Built binary PD classification models using structured financial and transactional data to reduce credit risk. Engineered features from tabular data (transaction frequency, recency, balance volatility) to improve signal quality.
- Implemented and benchmarked logistic regression, random forest, XGBoost, and LightGBM; selected best model via KS statistic and ROC-AUC, reducing false negative rate on defaults by 18%.
- Evaluated robustness with ROC-AUC, precision–recall curves, and KS across train/validation/test splits to prevent overfitting.
- Addressed class imbalance using SMOTE and asymmetric weighting, improving default recall without reducing overall precision.

IT Specialist

May 2023 – July 2025

- Applied hardware and software updates proactively, which reduced downtime by 20%.

University of South Florida

Undergraduate Research Assistant

Tampa, Florida

September 2025 – November 2025

- Developed and evaluated neural network models for protein disorder prediction, analyzing generalization behavior across multiple benchmark sequence datasets.
- Designed a systematic evaluation framework to benchmark model performance across varied input distributions, enabling reproducible comparison across architectures.
- Diagnosed failure modes in neural network predictions by comparing model outputs against simulation ground truth, identifying systematic bias in out-of-distribution inputs.

EtaFlex LLC

Software Engineer Intern

Santa Clara, California

June 2025 – August 2025

- Implemented backend data processing pipeline in Python handling high-throughput requests.
- Designed RESTful APIs enabling structured communication between microservices, supporting scalable backend architecture.

PROJECTS

Algorithmic Trading Backtester | Python, XGBoost, Streamlit • [GitHub](#) • [Live Demo](#)

- Built production backtesting framework for 3 quant strategies on 20 S&P 500 stocks (2015–2024) with structural lookahead-bias prevention and 75-test suite.
- Momentum strategy achieved Sharpe 1.30 and 30% annualized alpha over SPY; walk-forward XGBoost signal model achieved Sharpe 1.12 with 14.93% alpha.
- Deployed interactive Streamlit dashboard with live parameter controls, 9 tabs, and cached ML inference across all three strategies.

EDUCATION

University of Chicago, Chicago, IL

Physical Sciences Division

Master of Science in Computer Science

Sep 2026 – Dec 2027

University of South Florida, Tampa, FL

Bellini College of Artificial Intelligence, Cybersecurity and Computing

Bachelor of Science in Computer Science

Magna Cum Laude

Aug 2022– May 2026

Honors & Awards

Government Merit Scholarship (2026), Dean's List USF (2023), Knight Hacks Winner (2023), USF Green & Gold Award (2022)

ADDITIONAL INFORMATION

Languages: Azerbaijani (Native), Russian (Native), English (Full Proficiency), Turkish (Full Proficiency)

Programming & Development: Python, C/C++, SQL

ML & Algorithms: scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, Logistic Regression, Random Forest, Neural Networks

Evaluation & Statistics: ROC-AUC, Precision-Recall, KS Statistic, SMOTE, Bias-Variance Analysis

Backend & Databases: Django, REST API, PostgreSQL, MySQL, MongoDB

Tools & Infrastructure: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Jupyter Notebook, OpenAI API, Docker, Git/GitHub, Linux, Streamlit

Specialization: Artificial Intelligence, Machine Learning, and Quantitative Computing